



DRAFT OGC GEOSCIENCE MARKUP LANGUAGE (GEOSCIML) - PART 3: JSON ENCODING

STANDARD
Implementation

DRAFT

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ABSTRACT

GeoSciML (Geoscience Markup Language) is a Standard for representing geological features commonly described and portrayed in geological maps, cross sections, geological reports and databases. The Standard is composed of multiple parts. Part 1 describes a logical model. Part 3 (this document) describes encoding rules for the exchange of geological map data, geological time scales, boreholes, and metadata for laboratory analyses. The encoding rules define an application of Features and Geometry JSON, also known as JSON-FG.



KEYWORDS

The following are keywords to be used by search engines and document catalogues.

ogcdoc, OGC document, JSON, JSON-FG, schema, GeoSciML, geology, geoscience



PREFACE

The primary goal of this specification is to capture the schema, and encoding syntax of key elements described and portrayed in geological maps and databases, in order to enable information systems to interoperate with such data.



SECURITY CONSIDERATIONS

No security considerations have been made for this Standard.



SUBMITTING ORGANIZATIONS

The following organizations submitted this Document to the Open Geospatial Consortium (OGC):

- Organization One
- Organization Two



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1 SCOPE

1

SCOPE

Part 3 of GeoSciML covers the representation in JSON of features from the domain of geology (earth materials, geological units and stratigraphy, geological time, geological structures, geomorphology, geochemistry) and sampling features common to the practice of geoscience, such as boreholes and geological specimens.



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CONFORMANCE

This standard defines a JSON encoding which conforms to the OGC JSON-FG 1.0 encoding rules.

Requirements for one standardization target type are considered:

- Data instance

The normative artefacts for JSON encoding are the JSON Schema files provided online with this specification. Those documents explicitly provide the requirements that must be met by any JSON instance document claiming compliance to this specification. Any requirements that cannot be expressed in JSON Schema are described in the relevant section of this document. Therefore, compliant JSON instances shall:

1. validate with JSON Schema files,
2. pass compliance tests listed in relevant compliance sections.

Conformance with this standard shall be checked using all the relevant tests specified in Annex A (normative) of this document. The framework, concepts, and methodology for testing, and the criteria to be achieved to claim conformance are specified in the [OGC Compliance Testing Policies and Procedures](#) and the [OGC Compliance Testing web site](#).

All requirements-classes and conformance-classes described in this document are owned by the standard(s) identified.



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NORMATIVE REFERENCES

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO: ISO/TS 19103:2005, *Geographic information – Conceptual schema language*. International Organization for Standardization, Geneva (2005). <https://www.iso.org/standard/37800.html>.

ISO: ISO 19107:2003, *Geographic information – Spatial schema*. International Organization for Standardization, Geneva (2003). <https://www.iso.org/standard/26012.html>.

ISO: ISO 19108:2002/Cor 1:2006, *Geographic information – Temporal schema – Technical Corrigendum 1*. International Organization for Standardization, Geneva (2002). <https://www.iso.org/standard/44883.html>.

ISO: ISO 8601, *Data elements and interchange formats – Information interchange – Representation of dates and times*. International Organization for Standardization, Geneva <https://www.iso.org/standard/40874.html>.

ISO: ISO 19156:2011, *Geographic information – Observations and measurements*. International Organization for Standardization, Geneva (2011). <https://www.iso.org/standard/32574.html>.

ISO: ISO 19111:2007, *Geographic information – Spatial referencing by coordinates*. International Organization for Standardization, Geneva (2007). <https://www.iso.org/standard/41126.html>.

ISO: ISO 19115:2003, *Geographic information – Metadata*. International Organization for Standardization, Geneva (2003). <https://www.iso.org/standard/26020.html>.

ISO: ISO 19101:2002, *Geographic information – Reference model*. International Organization for Standardization, Geneva (2002). <https://www.iso.org/standard/26002.html>.

Simon Cox: OGC 10-025r1, *Observations and Measurements – XML Implementation*. Open Geospatial Consortium (2011). <https://docs.ogc.org/is/10-025r1/10-025r1/pdf>.

T. Berners-Lee, R. Fielding, L. Masinter: IETF RFC 3986, *Uniform Resource Identifier (URI): Generic Syntax*. RFC Publisher (2005). <https://www.rfc-editor.org/info/rfc3986>.

OMG: Unified Modeling Language (UML). Version 2.3

INSPIRE Thematic Working Group Geology: *INSPIRE Data Specification for the spatial data theme Geology Version 3.0*. European Commission Joint Research Centre (2013)

Clemens Portele, Panagiotis (Peter) A. Vretanos: OGC 21-045r1, *OGC Features and Geometries JSON – Part 1: Core*. Open Geospatial Consortium (2026). <http://www.opengis.net/doc/IS/json-fg-1/1.0>.

ISO: ISO 19101-1:2014, *Geographic information – Reference model – Part 1: Fundamentals*. International Organization for Standardization, Geneva (2014). <https://www.iso.org/standard/59164.html>.



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TERMS AND DEFINITIONS

This document uses the terms defined in [OGC Policy Directive 49](#), which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word “shall” (not “must”) is the verb form used to indicate a requirement to be strictly followed to conform to this document and OGC documents do not use the equivalent phrases in the ISO/IEC Directives, Part 2.

This document also uses terms defined in the OGC Standard for Modular specifications (OGC 08-131r3), also known as the ‘ModSpec’. The definitions of terms such as standard, specification, requirement, and conformance test are provided in the ModSpec.

For the purposes of this document, the following additional terms and definitions apply.

4.1. feature

abstraction of real world phenomena

Note 1 to entry: A feature may occur as a type or an instance. Feature type or feature instance shall be used when only one is meant.

[SOURCE: ISO 19101-1:2014]

4.2. feature collection

set of features from a dataset

[SOURCE: ISO 19168-1:2020]

4.3. JSON document

an information resource (series of octets) described by the “application/json” media type [JSON Schema 2020-12]

Note 1 to entry: The terms “JSON document,” “JSON text,” and “JSON value” are interchangeable.

[SOURCE: OGC 21-045r1]

4.4. JSON object

an unordered set of name/value pairs mapping the name to a JSON value [JSON Schema 2020-12, modified]

4.5. JSON-FG document

a JSON document that contains a JSON-FG root object

[SOURCE: OGC 21-045r1]

4.6. JSON-FG feature

a JSON object that validates against the JSON-FG feature schema

[SOURCE: OGC 21-045r1]

4.7. JSON-FG feature collection

a JSON object that validates against the JSON-FG feature collection schema

[SOURCE: OGC 21-045r1]

4.8. JSON-FG geometry

a JSON object that validates against the JSON-FG geometry schema

[SOURCE: OGC 21-045r1]

4.9. JSON-FG object

a JSON object that is either a JSON-FG feature, a JSON-FG feature collection, or a JSON-FG geometry

[SOURCE: OGC 21-045r1]

4.10. JSON-FG root object

a JSON-FG object that is not a direct children of another JSON-FG object

[SOURCE: OGC 21-045r1]

4.11. <JSON> key

the name of a member

[SOURCE: OGC 21-045r1]

4.12. <JSON> member

a name/value pair in a JSON object

[SOURCE: OGC 21-045r1]



5

CONVENTIONS

This sections provides details and examples for any conventions used in the document. Examples of conventions are symbols, abbreviations, use of JSON schema, or special notes regarding how to read the document.

5.1. Identifiers ---

The normative provisions in this specification are denoted by a URI constructed using this pattern:

<http://www.opengis.net/spec/{standard}/{m.n}>

All requirements and conformance tests that appear in this document are denoted by a partial URI which is relative to this base. The identifier supports cross-referencing of class membership, dependencies, and links from each conformance test to the requirements tested. In this specification identifiers are expressed as partial URIs or paths, which can be appended to a base URI that identifies the specification as a whole in order to construct a complete URI for identification in an external context.

The URI for each requirements class has the form:

[http://www.opengis.net/spec/geosciml-part2/5.0/req/\[classM\]](http://www.opengis.net/spec/geosciml-part2/5.0/req/[classM])

The URI for each requirement or recommendation has the form:

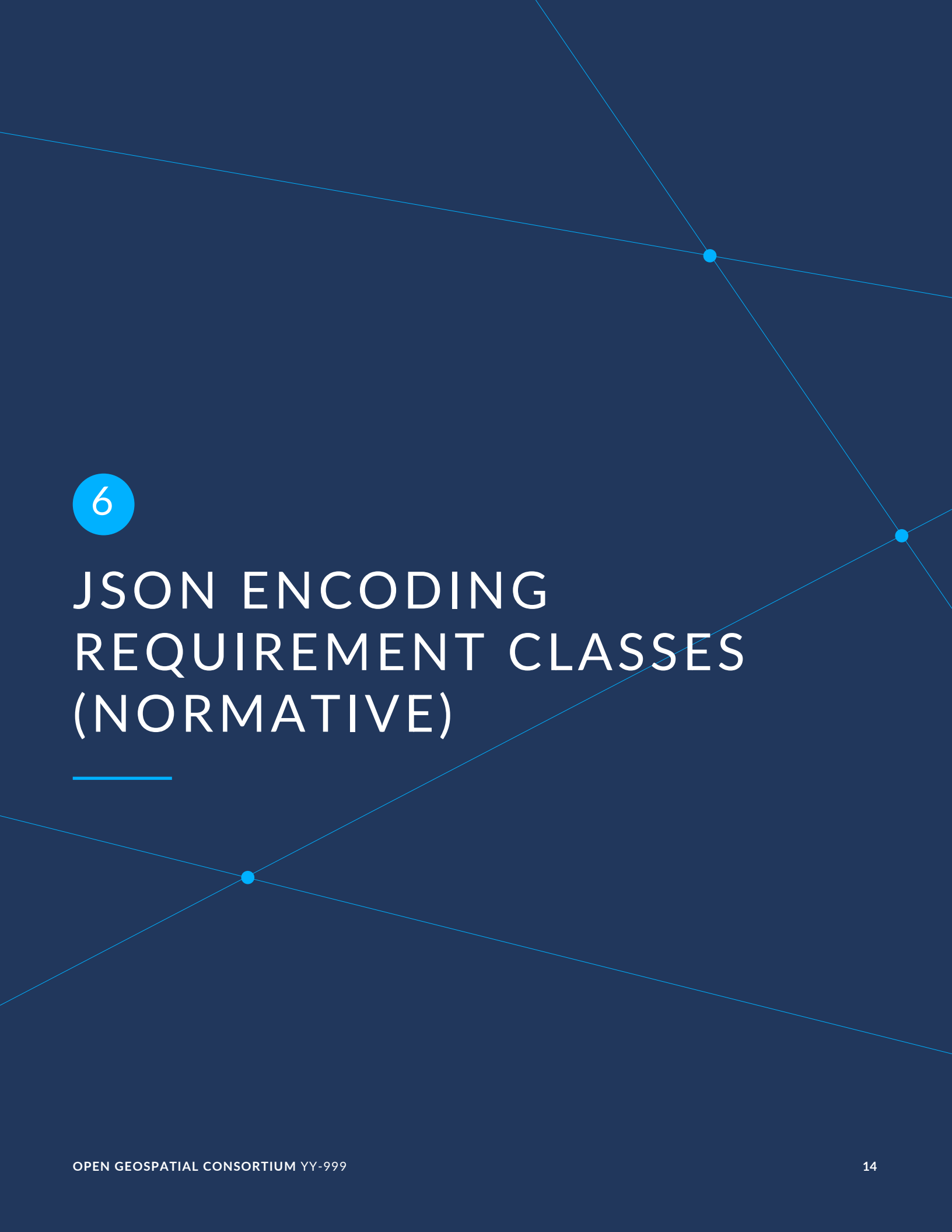
[http://www.opengis.net/spec/geosciml-part2/5.0/req/\[classM\]/\[reqN\]](http://www.opengis.net/spec/geosciml-part2/5.0/req/[classM]/[reqN])

The URI for each conformance class has the form:

[http://www.opengis.net/spec/geosciml-part2/5.0/conf/\[classM\]](http://www.opengis.net/spec/geosciml-part2/5.0/conf/[classM])

The URI for each conformance test has the form:

[http://www.opengis.net/spec/geosciml-part2/5.0/req/\[classM\]/\[testN\]](http://www.opengis.net/spec/geosciml-part2/5.0/req/[classM]/[testN])



6

JSON ENCODING REQUIREMENT CLASSES (NORMATIVE)

6

JSON ENCODING REQUIREMENT CLASSES (NORMATIVE)

JSON Schema files were derived from the UML model following the [OGC Best Practice for UML to JSON Encoding Rules \(OGC 24-017r1\)](#).

NOTE: The SWG should replace this with a diagram from Enterprise Architect

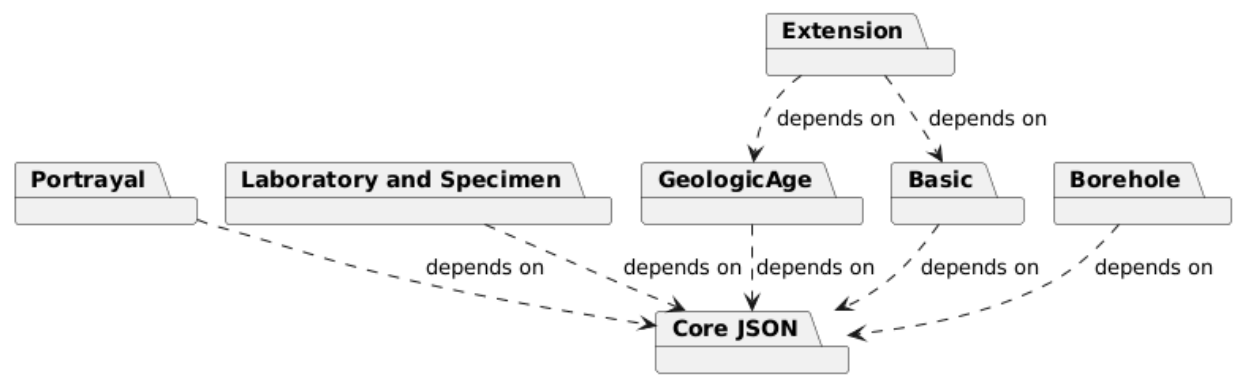


Figure 1 – JSON Encoding requirements classes dependencies (external dependencies not shown)

6.1. GeoSciML Core JSON Abstract Requirements Class (Normative)

REQUIREMENTS CLASS 1: REQUIREMENTS CLASS 'GEOSCIML CORE JSON'	
IDENTIFIER	https://www.opengis.net/spec/geosciml-part3/5.0/req/gsmlljson-core
TARGET TYPE	Data instance
	Web API
	https://www.opengis.net/spec/geosciml-part1/5.0/req/gsmll-core
	JSON Schema
PREREQUISITES	ISO 19118:2011 Geographic information – Encoding
	OGC Features and Geometries JSON – Part 1: Core v1.0.0
	OGC Abstract Specification Topic 20: Observations, measurements and samples v3.0.0
	OGC SWE Common Data Model Encoding Standard v3.0.0, Clause 9

REQUIREMENTS CLASS 1: REQUIREMENTS CLASS 'GEOSCI ML CORE JSON'

NORMATIVE STATEMENTS

Recommendation 1: /rec/gsmljson-core/identifier-uri
Recommendation 2: /rec/gsmljson-core/iso8601-time
Requirement 1: /req/gsmljson-core/json
Requirement 2: /req/gsmljson-core/codelist
Requirement 4: /req/gsmljson-core/time-zone

This requirements class is shared by all JSON GeoSciML instances.

6.1.1. JSON document validation

A JSON instance shall validate to JSON encoding rules provided by this specification for each of the JSON requirements classes.

REQUIREMENT 1

IDENTIFIER /req/gsmljson-core/json

INCLUDED IN Requirements class 1: <https://www.opengis.net/spec/geosciml-part3/5.0/req/gsmljson-core>

A An JSON instance document SHALL validate against the JSON Schema files provided in this Standard.

6.1.2. CodeList

Open code lists (see 8.2.7) are encoded simply as an HTTP URI (or HTTPS URI). See the following example that references the 'mudstone' term from a code list.

```
"material": {
  "type": "Feature",
  "featureType": "RockMaterial",
  "id": "hervey-group-mudstone",
  "geometry": null,
  "properties": {
    "lithology": "http://resource.geosciml.org/classifier/cgi/
simplelithology/mudstone"
  }
}
```

The URI should resolve to a resource representation (often a SKOS document in Turtle or JSON-LD). The resource can have multiple representations and it is not guaranteed that a JSON-LD parsable document can be obtained from the vocabulary service.

REQUIREMENT 2

IDENTIFIER	/req/gsmljson-core/codelist
INCLUDED IN	Requirements class 1: https://www.opengis.net/spec/geosciml-part3/5.0/req/gsmljson-core
A	Vocabulary term SHALL be encoded with an HTTP/HTTPS URI that is resolvable to the definition of that term.

6.1.3. Identifiers

To enable consistent handling of identifiers in GeoJSON and JSON-FG applications, the `id` member of a GeoJSON feature is used to encode the identifier of a GeoSciML feature.

REQUIREMENT 3

IDENTIFIER	/req/gsmljson-core/identifier
A	The <code>id</code> member of a GeoJSON feature shall be used to encode the identifier of a GeoSciML feature.

RECOMMENDATION 1

IDENTIFIER	/rec/gsmljson-core/identifier-uri
INCLUDED IN	Requirements class 1: https://www.opengis.net/spec/geosciml-part3/5.0/req/gsmljson-core
A	Feature identifiers (unique name) provided in the <code>'id'</code> property of a feature SHOULD be URI of resource using Linked Open Data principles.

```
{
  "type": "Feature",
  "id": "http://example.org/feature/asc/geologicunit/stratno/25947",
  "featureType": "GeologicUnit",
  "geometry": null,
  ...
}
```

6.1.4. Nillables or Voidables

A nillable property (identified as “voidable” in the UML model) is a property that can document the reason that a value is not provided. There are two ways to identify a nil value.

- Using XSD `xsi:nil="true"` and `nilReason`. Where available, this encoding method should be used to indicate nilled property values.
- Using a HTTP URI identifier defined by a community as representing a null value. OGC uses <http://www.opengis.net/def/nil/OGC/0/{nilReason}> (OGC 12-110).

6.1.5. Date encoding

The date-time values shall conform to ISO 8601 standards. Although this is already a GML 3.2 encoding rule (clause 14.2.2.7), this format should also be used in any string that should contain a date, or date and time.

RECOMMENDATION 2

IDENTIFIER /rec/gsmljson-core/iso8601-time

INCLUDED IN Requirements class 1: <https://www.opengis.net/spec/geosciml-part3/5.0/req/gsmljson-core>

A All date-time occurrences SHOULD be encoded using ISO8601 extended time format.

Note that this precludes the use of time-coordinate systems such as UNIX time. This is specified in order to be maximally consistent with TimeSeriesML requirements.

The time zone shall be included in the time element.

REQUIREMENT 4

IDENTIFIER /req/gsmljson-core/time-zone

INCLUDED IN Requirements class 1: <https://www.opengis.net/spec/geosciml-part3/5.0/req/gsmljson-core>

A The value of each time element SHALL include a time zone definition using a signed 4 digit character or a 'Z' to represent Zulu or Greenwich Mean Time (GMT). This is defined by the following regular expression:
(Z|[+-]HH:MM).

Greenwich Mean Time (GMT or Zulu)

```

---
<om:phenomenonTime>
  <gml:TimeInstant gml:id="ti.1">
    <gml:timePosition>1981-09-12T00:00:00Z</gml:timePosition>
  </gml:TimeInstant>
</om:phenomenonTime>
---
```

Time Zone (example is Newfoundland time zone -3:30)

```
---
<om:phenomenonTime>
  <gml:TimeInstant gml:id="ti.2">
    <gml:timePosition>1981-09-12T00:00:00-03:30</gml:timePosition>
  </gml:TimeInstant>
</om:phenomenonTime>
---
```

6.2. GeoSciML Basic JSON Requirements Class (Normative)

REQUIREMENTS CLASS 2: REQUIREMENTS CLASS 'GEOSCI ML BASIC JSON'

IDENTIFIER	https://www.opengis.net/spec/geosciml-part3/5.0/req/gsmljson-basic
TARGET TYPE	Data instance
PREREQUISITES	https://www.opengis.net/spec/geosciml-part1/5.0/req/gsml-basic/req/gsmljson-core
NORMATIVE STATEMENT	Requirement 5: /req/gsmljson-basic/json

All the elements from the basic package must be schema valid according to the XSD document provided at <http://schemas.opengis.net/gsml/4.1/geoSciMLBasic.xsd>

REQUIREMENT 5

IDENTIFIER	/req/gsmljson-basic/json
INCLUDED IN	Requirements class 2: https://www.opengis.net/spec/geosciml-part3/5.0/req/gsmljson-basic
A	A JSON instance document SHALL validate with schema located at http://schemas.opengis.net/gsml/4.1/geoSciMLBasic.xsd .

All the elements from basic package must pass the schematron rules defined in the schematron file located at <http://schemas.opengis.net/gsml/4.1/geoSciMLBasic.sch>.

Some properties links to stub property blocks (see 5.1) whose values are empty abstract classes in the GeoSciML Basic package. Since abstract description classes are DataType, the property is inline only and therefore an empty property is not schema valid.

This instance is **not** XSD valid:

```
<?xml version="1.0" encoding="UTF-8"?>
<GeologicUnit xmlns="http://www.opengis.net/gsml/4.1/GeoSciML-Basic"
  xmlns:gml="http://www.opengis.net/gml/3.2" gml:id="x1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://schemas.opengis.net/GeoSciML-Basic/4.1
    http://schemas.opengis.net/geosciml/4.1/geoSciMLBasic.xsd">
...
    <gbMaterialDescription></gbMaterialDescription>
...
</GeologicUnit>
```

6.2.1. relatedFeature

In GeoSciML Basic, relatedFeature is byReference only. AbstractFeatureRelation subtypes are materialized in Extension package.

6.3. GeoSciML Extension JSON Requirements Class (Normative)

REQUIREMENTS CLASS 3: REQUIREMENTS CLASS 'GEOSCIML EXTENSION JSON'

IDENTIFIER	https://www.opengis.net/spec/geosciml-part3/5.0/req/gsmljson-extension
TARGET TYPE	Data instance
PREREQUISITES	https://www.opengis.net/spec/geosciml-part1/5.0/req/gsml-extension/req/gsmljson-basic /req/gsmljson-geologictime
NORMATIVE STATEMENT	Requirement 6: /req/gsmljson-extension/json

All the elements from the GeoSciML Extension package must be schema valid according to the XSD document provided at <http://schemas.opengis.net/gsml/4.1/geoSciMLExtension.xsd>.

REQUIREMENT 6

IDENTIFIER [/req/gsmljson-extension/json](https://www.opengis.net/spec/geosciml-part3/5.0/req/gsmljson-extension/json)

REQUIREMENT 6

INCLUDED IN	Requirements class 3: https://www.opengis.net/spec/geosciml-part3/5.0/req/gsmljson-extension
A	A JSON instance document SHALL validate with schema located at http://schemas.opengis.net/gsml/4.1/ geoSciMLEExtension.xsd .

All the elements from the GeoSciML Extension package must pass the schematron rules defined in the schematron file located at <http://schemas.opengis.net/gsml/4.1/geoSciMLEExtension.sch>.

6.4. GeoSciML GeologicTime JSON Requirements Class (Normative)

REQUIREMENTS CLASS 4: REQUIREMENTS CLASS 'GEOSCIML GEOLOGICTIME JSON'

IDENTIFIER	https://www.opengis.net/spec/geosciml-part3/5.0/req/gsmljson-geologictime
TARGET TYPE	Data instance
PREREQUISITES	https://www.opengis.net/spec/geosciml-part1/5.0/req/gsml-geologictime/req/gsmljson-core
NORMATIVE STATEMENT	Requirement 7: /req/gsmljson-geologictime/json

All the elements from the GeoSciML Geologic Time package must be schema valid according to the XSD document provided at <http://schemas.opengis.net/gsml/4.1/geologicTime.xsd>.

REQUIREMENT 7

IDENTIFIER	/req/gsmljson-geologictime/json
INCLUDED IN	Requirements class 4: https://www.opengis.net/spec/geosciml-part3/5.0/req/gsmljson-geologictime
A	A JSON instance document SHALL validate with schema located at http://schemas.opengis.net/gsml/4.1/geologicTime.xsd .

All the elements from the GeoSciML Geologic Time package must pass the schematron rules defined in the schematron file located at <http://schemas.opengis.net/gsml/4.1/geologictime.sch>.

6.5. GeoSciML Borehole JSON Requirements Class (Normative)

REQUIREMENTS CLASS 5: REQUIREMENTS CLASS 'GEOSCIML BOREHOLE JSON'

IDENTIFIER	https://www.opengis.net/spec/geosciml-part3/5.0/req/gsmljson-borehole
TARGET TYPE	Data instance
PREREQUISITES	https://www.opengis.net/spec/geosciml-part1/5.0/req/gsml-borehole/req/gsmljson-core
NORMATIVE STATEMENT	Requirement 8: /req/gsmljson-borehole/json

All the elements from the Borehole package must be schema valid according to the XSD document provided at <http://schemas.opengis.net/gsml/4.1/borehole.xsd>.

REQUIREMENT 8

IDENTIFIER	/req/gsmljson-borehole/json
INCLUDED IN	Requirements class 5: https://www.opengis.net/spec/geosciml-part3/5.0/req/gsmljson-borehole
A	A JSON instance document shall validate with schema located at http://schemas.opengis.net/gsml/4.1/borehole.xsd .

All the elements from the Borehole package must pass the schematron rule defined in the schematron file located at <http://schemas.opengis.net/gsml/4.1/borehole.sch>.

6.6. GeoSciML LaboratoryAnalysis-Specimen JSON Requirements Class (Normative)

REQUIREMENTS CLASS 6: REQUIREMENTS CLASS 'GEOSCIML LABORATORYANALYSIS-SPECIMEN JSON'

IDENTIFIER	https://www.opengis.net/spec/geosciml-part3/5.0/req/gsmljson-lab
------------	---

REQUIREMENTS CLASS 6: REQUIREMENTS CLASS 'GEOSCI ML LABORATORYANALYSIS-SPECIMEN JSON'

TARGET TYPE	Data instance
PREREQUISITES	/req/gsml-lab /req/gsmljson-core
NORMATIVE STATEMENT	Requirement 9: /req/gsmljson-lab/json

All the elements from the GeoSciML Laboratory Analysis & Specimen package must be schema valid according to the XSD document provided at <http://schemas.opengis.net/gsml/4.1/laboratoryAnalysis-Specimen.xsd>.

REQUIREMENT 9

IDENTIFIER	/req/gsmljson-lab/json
INCLUDED IN	Requirements class 6: https://www.opengis.net/spec/geosciml-part3/5.0/req/gsmljson-lab
A	A JSON instance document SHALL validate with schema located at http://schemas.opengis.net/gsml/4.1/laboratoryAnalysis-Specimen.xsd .

All the elements from the GeoSciML Laboratory Analysis & Specimen package must pass the schematron rule defined in the schematron file located at <http://schemas.opengis.net/gsml/4.1/laboratoryAnalysis-Specimen.sch>.

6.7. Abstract GeoSciML Lite JSON Requirements Class (Normative)

REQUIREMENTS CLASS 7: REQUIREMENTS CLASS 'ABSTRACT GEOSCI ML LITE JSON'

IDENTIFIER	https://www.opengis.net/spec/geosciml-part3/5.0/req/gsmljson-lite
TARGET TYPE	Data instance
PREREQUISITES	https://www.opengis.net/spec/geosciml-part1/5.0/req/gsml-lite /req/gsmljson-core GML Simple Feature OGC 10-100r3 SF-0
NORMATIVE STATEMENTS	Recommendation 7-1: /rec/gsmljson-lite/SF-0-simpletype Recommendation 7-2: /rec/gsmljson-lite/user-json

REQUIREMENTS CLASS 7: REQUIREMENTS CLASS 'ABSTRACT GEOSCIML LITE JSON'

Recommendation 7-3: /rec/gsmljson-lite/user-ns

Requirement 10: /req/gsmljson-lite/SF-0

The abstract GeoSciML Lite encoding sets general encoding rules for JSON targets, regardless of version of JSON-FG.

6.7.1. Simple Feature

GeoSciML Lite schemas are meant to deliver simple content, consistent with simple scenarios described in GML Simple Feature Level 0 (OGC 10-100r3). User defined properties must respect the same constraints as defined in SF-0 specification.

REQUIREMENT 10

IDENTIFIER /req/gsmljson-lite/SF-0

INCLUDED IN Requirements class 7: <https://www.opengis.net/spec/geosciml-part3/5.0/req/gsmljson-lite>

A A JSON instance document SHALL be compliant to GML Simple Feature Level 0.

Simple Feature defines 3 compliance levels summarized in Table 17.

Table 1 — Reproduction of Table 1 of 10-100r3 (Clause 2.1)

	LEVEL SF-0	LEVEL SF-1	LEVEL SF-2
restricted set of built-in nonspatial property types	Yes ¹	Yes ¹	No
Restricted set of property-types	Yes ²	Yes ²	Yes ²
user-defined property types	No	Yes	Yes
use of nillable and xsi:nil	No	Yes	Yes
cardinality of properties	0..1	0..unbounded	0..unbounded
non-spatial property values references	Yes ³	Yes ³	Yes
spatial property values references	Yes ³	Yes ³	Yes

	LEVEL SF-0	LEVEL SF-1	LEVEL SF-2
1.	string, integer, measurement, date, real, binary, boolean, URI		
2.	Point, Curve (LineString), Surface (Polygon), Geometry, MultiPoint, MultiCurve, MultiSurface, MultiGeometry		
3.	In levels 0 and 1, remote values for properties are supported only through the use of the type gml:ReferenceType. The more generalized GML property-type pattern allowing mixed inline and byReference encoded property values within the same instance document is disallowed.		

6.8. GeoSciML Lite GML 3.1 profile (Normative)

REQUIREMENTS CLASS 8: REQUIREMENTS CLASS 'GEOSCIML LITE GML 3.1 PROFILE'

IDENTIFIER	https://www.opengis.net/spec/geosciml-part3/5.0/req/gsmljson-lite-31
TARGET TYPE	Data instance
PREREQUISITES	https://www.opengis.net/spec/geosciml-part1/5.0/req/gsml-lite/req/gsmljson-lite
NORMATIVE STATEMENT	Requirement 11: /req/gsmljson-lite-31/json

Because of the limited availability of WFS 2.0 compliant servers and clients, GeoSciML SWG officially supports WFS 1.1.0 and GML 3.1.1 for delivery of Lite features.

All the elements from the GeoSciML Lite package must be schema valid according to the XSD document provided at <http://schemas.opengis.net/gsml/4.1/geosciml-lite.xsd>.

REQUIREMENT 11

IDENTIFIER	/req/gsmljson-lite-31/json
INCLUDED IN	Requirements class 8: https://www.opengis.net/spec/geosciml-part3/5.0/req/gsmljson-lite-31
A	A JSON instance document shall validate with schema located at http://schemas.opengis.net/gsml/4.1/geosciml-portrayal.xsd .



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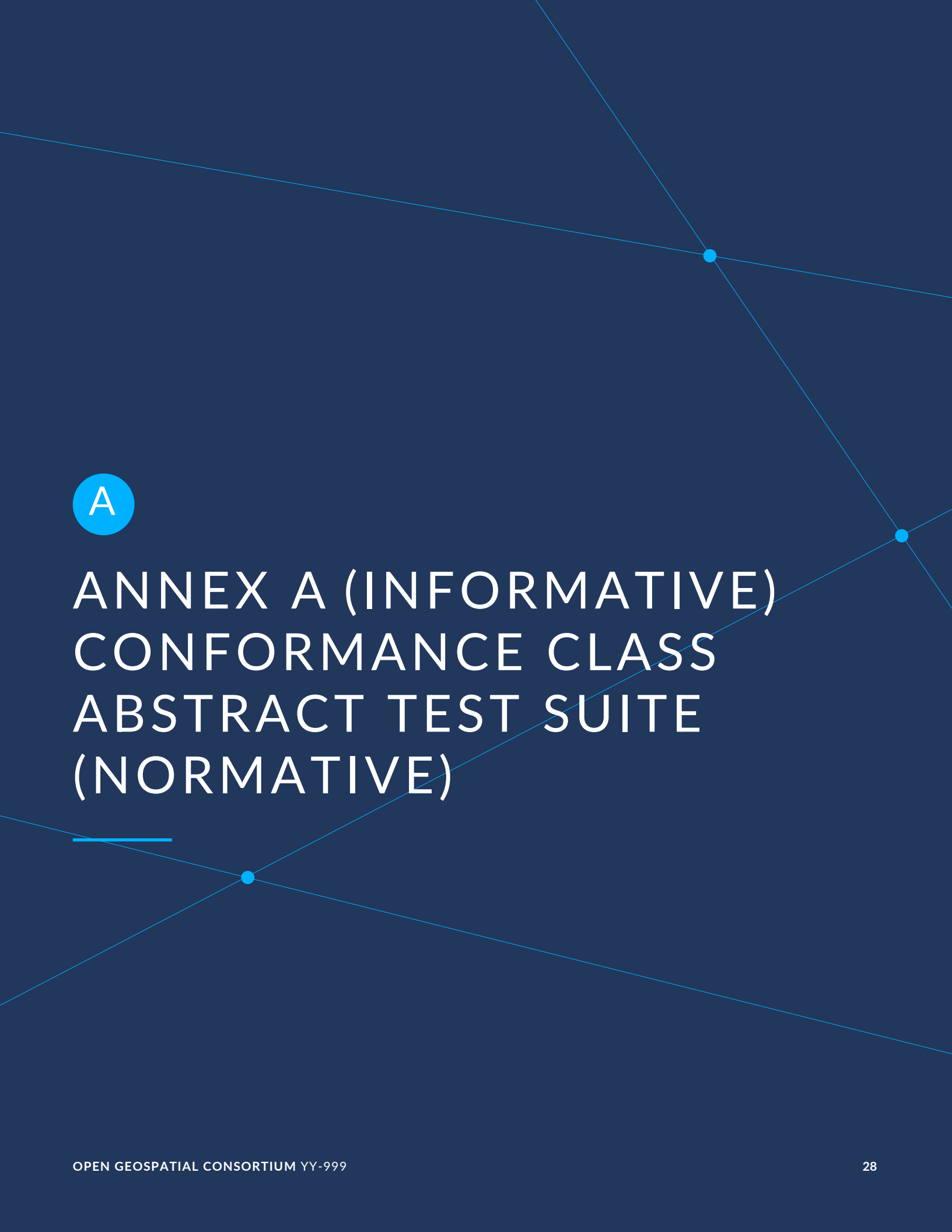
MEDIA TYPES FOR ANY DATA ENCODING(S)

7

MEDIA TYPES FOR ANY DATA ENCODING(S)

GeoSciML 5.0 data conforming to the logical model defined in Part 1 of GeoSciML can be encoded in documents conforming to JSON-FG. The standard MIME-type and sub-type for JSON-FG data should be used to indicate the encoding in internet exchange, as specified in OGC 21-045r1, namely

`application/geo+json`



A

ANNEX A (INFORMATIVE) CONFORMANCE CLASS ABSTRACT TEST SUITE (NORMATIVE)



ANNEX A

(INFORMATIVE)

CONFORMANCE CLASS ABSTRACT TEST SUITE (NORMATIVE)

NOTE: Ensure that there is a conformance class for each requirements class and a test for each requirement (identified by requirement name and number)

A.1. Conformance Class A

CONFORMANCE CLASS A.1	
IDENTIFIER	http://www.opengis.net/spec/name-of-standard/1.0/conf/example1
REQUIREMENTS CLASS	http://www.opengis.net/spec/ABCD/m.n/req/req-class-a
TARGET TYPE	Web API
CONFORMANCE TESTS	Abstract test A.1: /conf/core/map-response Abstract test A.2: /conf/core/http

A.1.1. Example 1

ABSTRACT TEST A.1	
IDENTIFIER	/conf/core/map-response
REQUIREMENT	/req/req-class-a/req-name-1
TEST PURPOSE	Verify that the implementation's response for the map retrieval operation is correct
TEST METHOD	A single step test method can be documented as a single line.

A.1.2. Example 2

ABSTRACT TEST A.2

IDENTIFIER /conf/core/http

REQUIREMENT /req/req-class-a/req-name-2

TEST PURPOSE Validate that the resource paths advertised through the API conform with HTTP 1.1 and, where appropriate, TLS.

TEST METHOD 1. A sequential multi-step test method can be documented as shown here. This is the first step.
 2. This is the second step.



ANNEX B (INFORMATIVE) TITLE



ANNEX B (INFORMATIVE) TITLE

NOTE: Place other Annex material in sequential annexes beginning with “B” and leave final two annexes for the Revision History and Bibliography



ANNEX C (INFORMATIVE) REVISION HISTORY



ANNEX C (INFORMATIVE) REVISION HISTORY

Table C.1

DATE	RELEASE	EDITOR	PRIMARY CLAUSES MODIFIED	DESCRIPTION
2016-04-28	0.1	G. Editor	all	initial version



BIBLIOGRAPHY





BIBLIOGRAPHY

- [1] ISO: ISO 19168-1:2020, *Geographic information – Geospatial API for features – Part 1: Core*. International Organization for Standardization, Geneva (2020). <https://www.iso.org/standard/32586.html>.